

Building Connected Application using WF and WCF

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Agenda

- Modeling processes via WF
- Exposing and consuming services via WCF

What is Workflow?

*The workflow framework and tools for
Microsoft products and
partner/customer ecosystem*

- Single workflow technology for Microsoft® Windows®
- A framework for building workflow, not an application or server
- Make declarative workflow a mainstream development paradigm

Why Workflow?

Real-World Examples

“Orders are confirmed in 48 hours and shipped within 30 days.”

“Most suppliers confirm our orders but some forget and we need to follow up.”

“What are the next steps in handling this order?”

Workflow Value-Add

Long Running and Stateful Workflows *run* for up to 30 days and maintain state throughout

Flexible Control Flow
Flexibility for people to override or skip steps in the workflow

Transparency
Rendering a visualization of next steps based on defined control flow

Architecture

Key Concepts

- Workflows are a set of Activities
- Workflows run within a Host Process: any application or server
- Developers can build their own Custom Activity Libraries

Components

- Base Activity Library: Out-of-box activities and base for custom activities
- Runtime Engine: Workflow execution and state management
- Runtime Services: Hosting flexibility and communication
- Visual Designer: Graphical and code-based construction



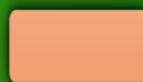
Extensible Activities

Base Activity Library



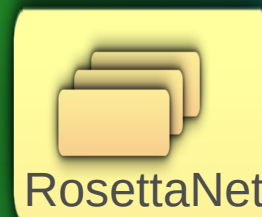
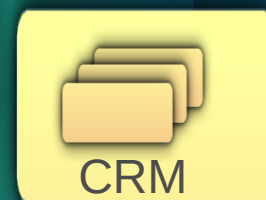
- OOB activities, workflow types, base types
- General-purpose
- Activity libraries define workflow constructs

Custom Activity Libraries



- Create/Extend/Compose activities
- App-specific building blocks
- First-class citizens

Domain-Specific Workflow Packages



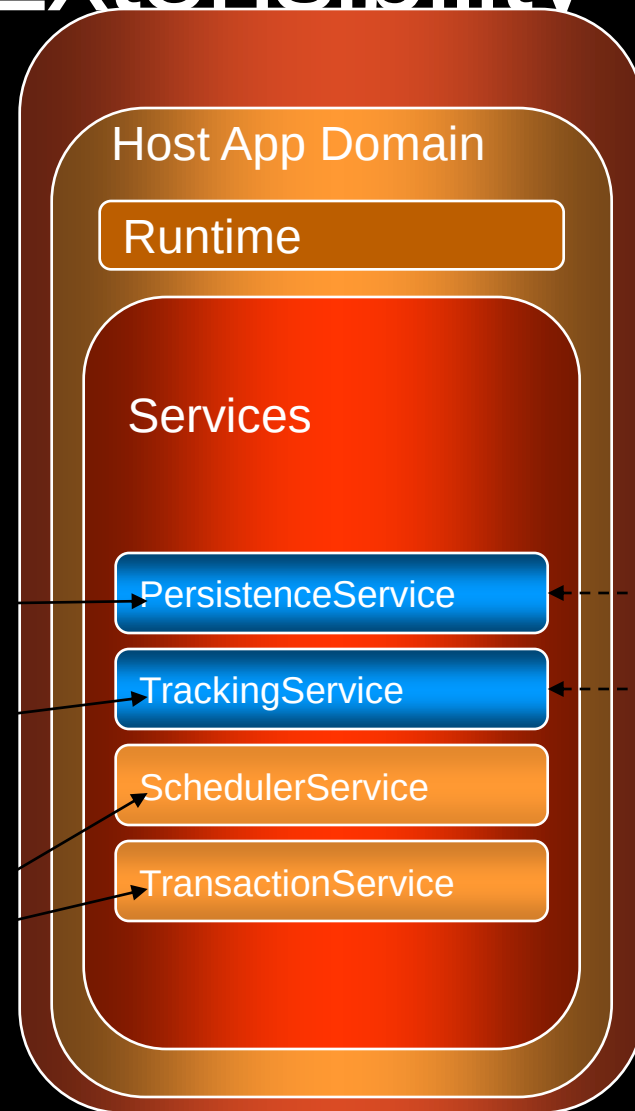
- Vertical-specific activities and workflows
- Best-practice intellectual property (IP) and knowledge

Runtime Extensibility

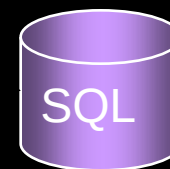
PersistenceService stores and retrieves instance state.

TrackingService manages profiles and stores tracked information.

Common resource services for managing threading, timers and creating transactions



Out of Box Services are provided that support SQL Server 2000 & 2005



Workflow Services

Advantages

- Service implemented as Workflow
 - Leverage Workflow development experience
 - Enforces application protocol by default
 - Allows long-running, durable Services
- Workflow hosted as Service
 - ServiceContract provides typed interface for integrating with the Workflow
 - Messages enforce by-value, serializable data
 - Workflows can be message-activated

Workflow Service

Architecture

Service.cs
Workflow.cs or
Workflow.xaml



App.config

Workflow
ServiceHost

ServiceDescription

WorkflowServiceBehavior

WorkflowOperationBehavior

Service Runtime

WorkflowOperationInvoker

OperationSelector

DurableInstanceProvider

MessageContextInspector

ContextChannel

ListenerChannel

Workflow Instance

ReceiveActivity 1

ReceiveActivity 2

Workflow Runtime

WF Persistence DB



Getting Started with WF

demo

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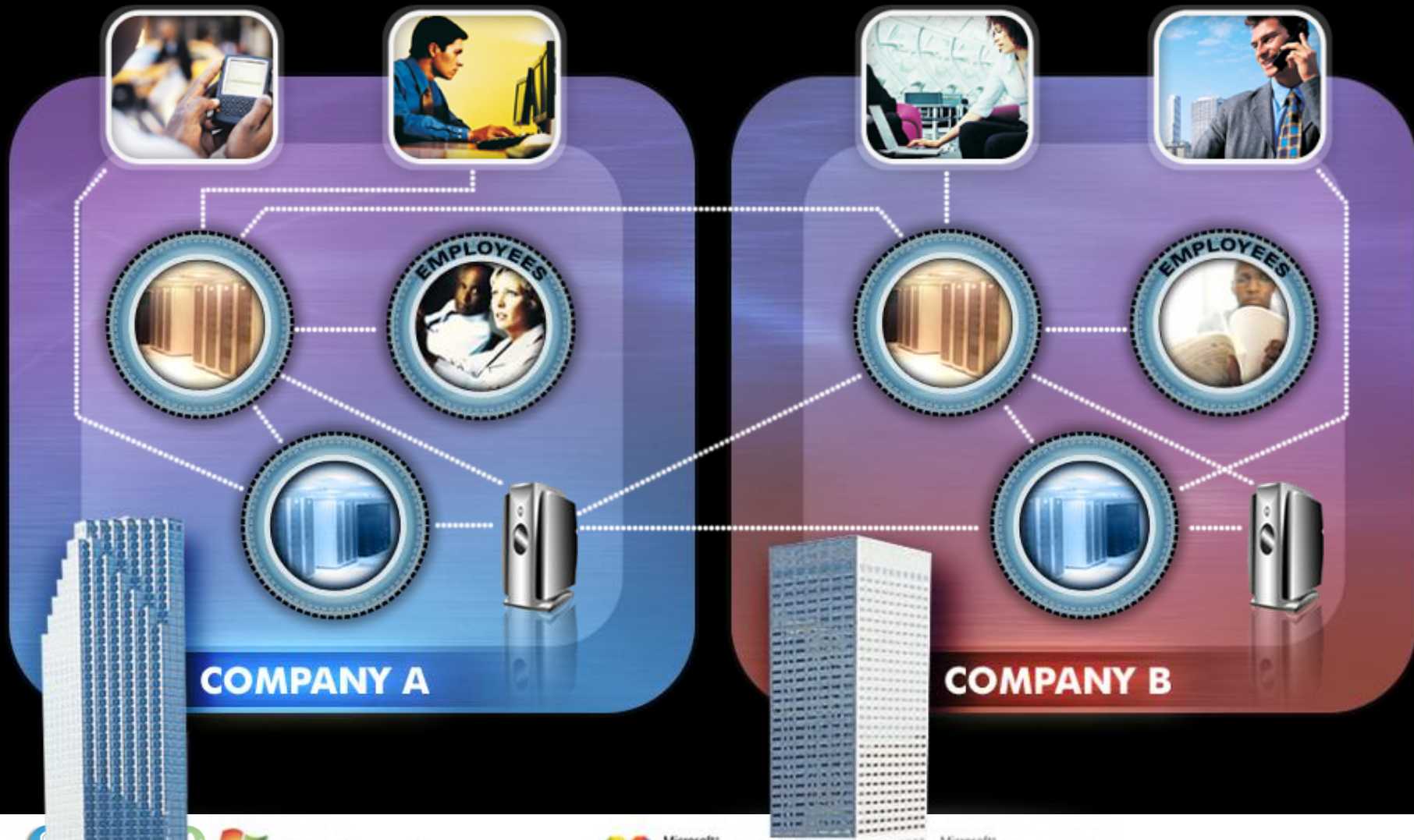
The Imperative to Connect

MOBILE EMPLOYEES

CUSTOMERS

CUSTOMERS

MOBILE EMPLOYEES

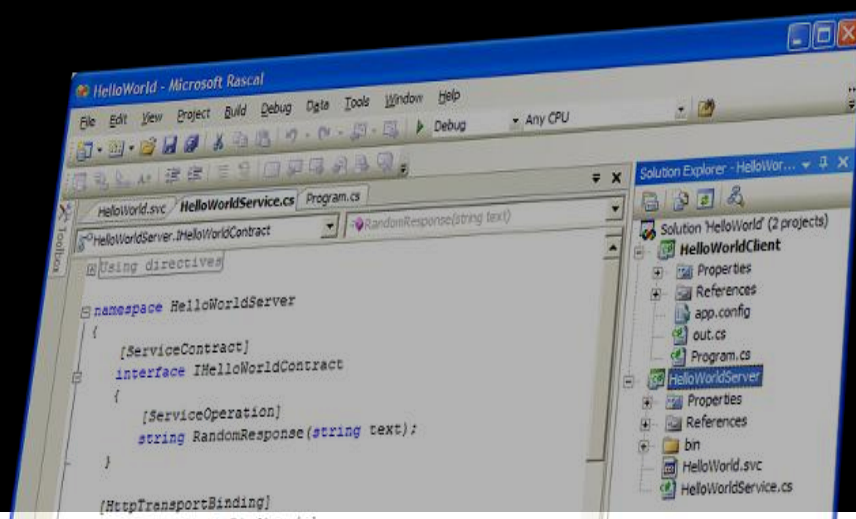


COMPANY A

COMPANY B

WCF is:

- Our next-generation platform for distributed systems
- A set of extensions to the Microsoft.NET Framework 2.0
- Build WCF services in Visual Studio 2008 using any .NET Language
 - Intelligent code editing, IDE Extensions for WCF, debugging, re-factoring, code snippets, etc.
 - Visual Basic .NET, Visual C#, etc.
- Runs on
 - Microsoft Windows Vista
 - Microsoft Windows XP
 - Microsoft Windows Server 2003
- Part of .NET 3.0 and .NET 3.5



Benefits of WCF

Unification

- Unifies today's distributed technologies
- Lets you develop/deploy/manage one model
- Visual Studio 2008 integration

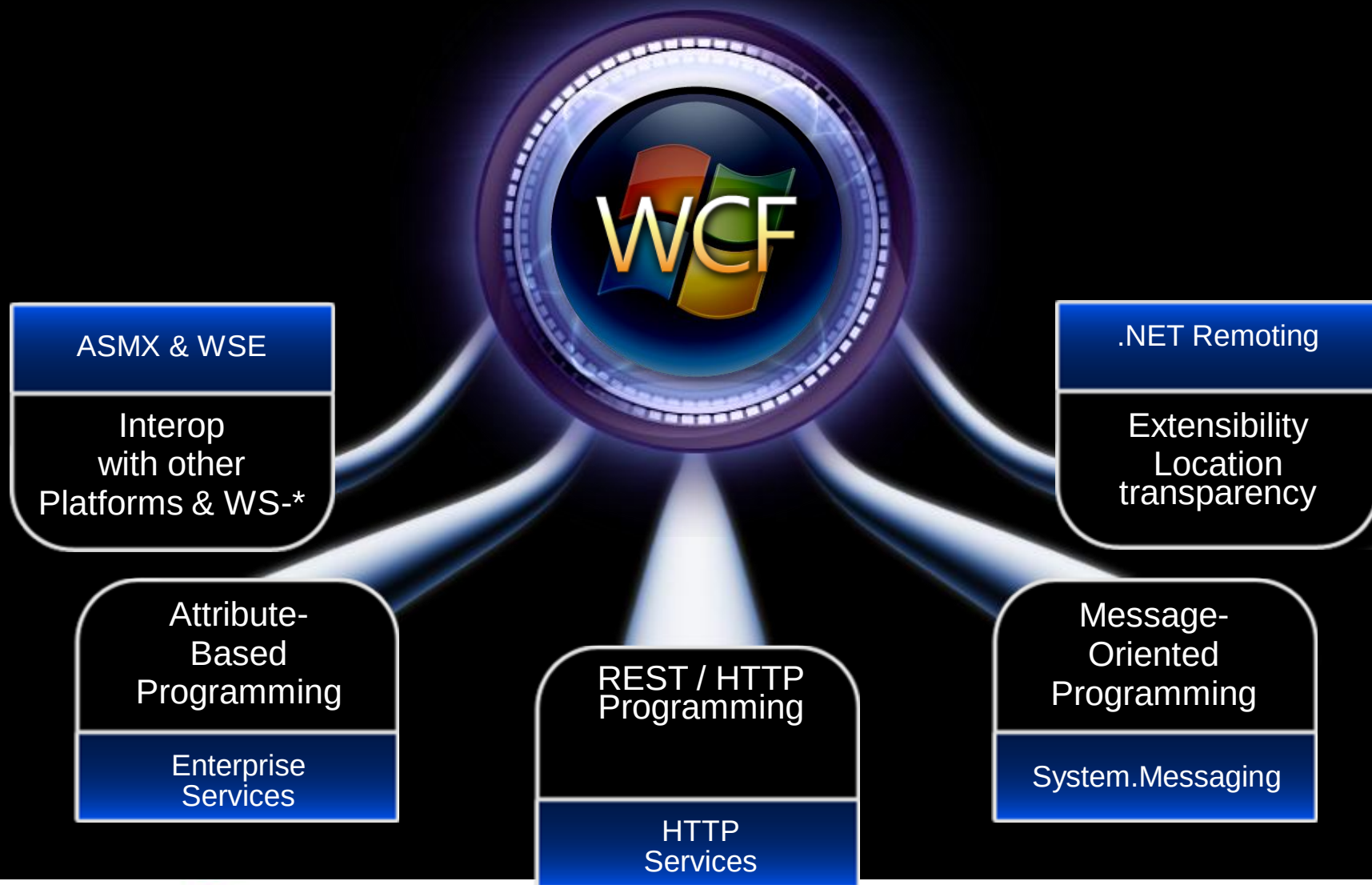
Interoperability

- Broad support for Web services (WS-*) specifications and REST
- Single stop shop for building services

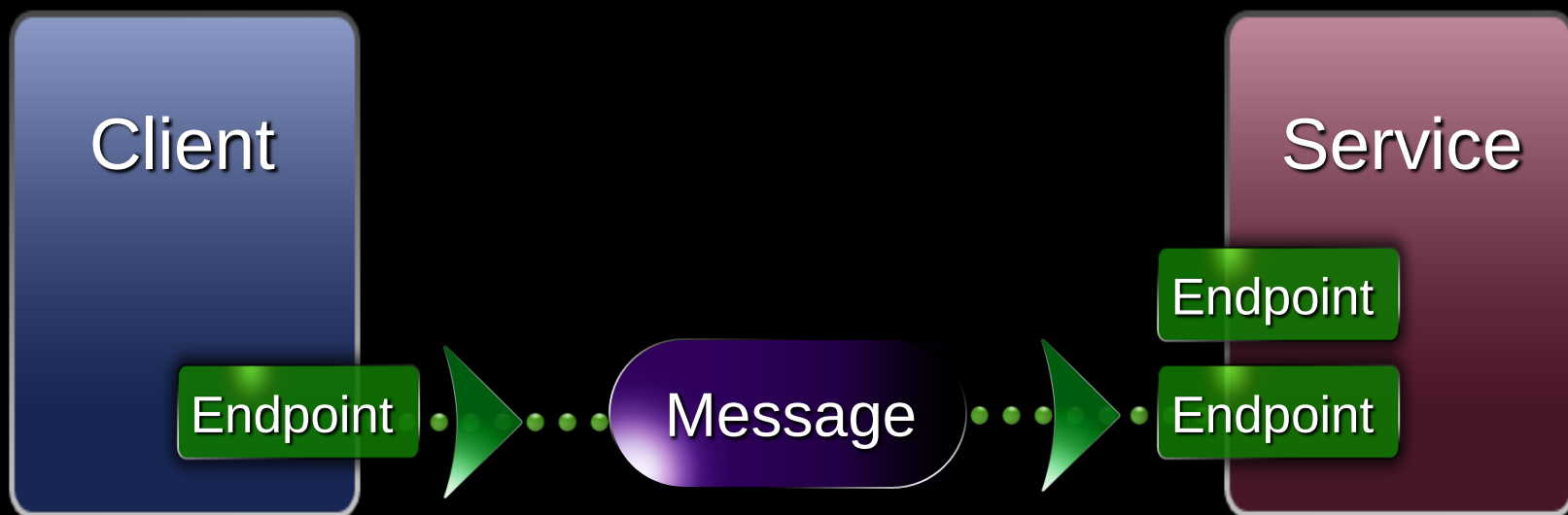
Service-Oriented Development

- Enables development of loosely-coupled services
- Config-based communication

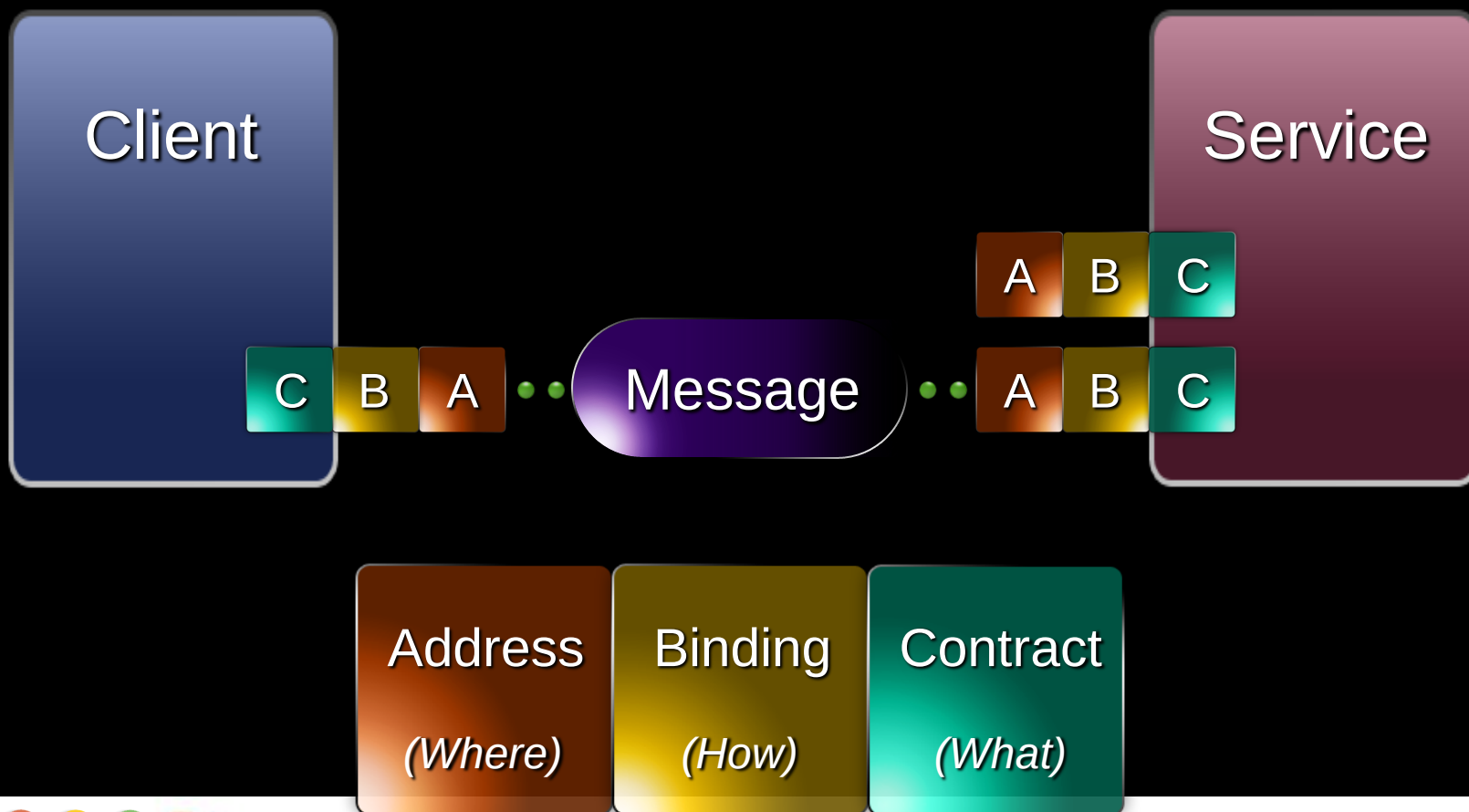
Unified Programming Model



A Look at WCF (1)



A Look at WCF (2)



Productivity

Using Visual Studio .NET 2003

```
class HelloService {  
  
    [WebMethod]  
    public String Hello(String Greeting) {  
  
        X509CertificateCollection collection = new  
        X509CertificateCollection();  
        IntPtr blob = Marshal.AllocHGlobal(Marshal.SizeOf(  
            typeof(CRYPTOAPI_BLOB)));  
        IntPtr data = (IntPtr)((int)blob +  
            Marshal.SizeOf(typeof(CRYPTOAPI_BLOB)));  
        ...  
  
        SeqAckRange range = new SeqAcknRange(id, low, high );  
        SeqAckRange[] ranges = { range };  
        ReliableQueue.ProcessAcks( ranges );  
        ...  
  
        BeginTransaction (NULL,  
            ISOLATIONLEVEL_SERIALIZABLE, 0, pITxOptions,  
            &pITransaction);  
        ...  
  
        return Greeting;  
    }  
}
```

20,379 lines
security

5,988 lines
reliable messaging

25,507 lines
transactions

4,442 lines
infrastructure

Total lines 56,296



Productivity

Using Visual Studio .NET 2003 and WSE

```
class HelloService {  
  
    [WebMethod]  
    public String Hello(String Greeting) {  
  
        foreach ( SecurityToken tok in  
            requestContext.Security.Tokens  
        {  
            X509SecurityToken token = tok as X509SecToken  
        }  
  
        ...  
        SeqAckRange range = new SeqAcknRange(id,low,high );  
        SeqAckRange[] ranges = { range };  
        ReliableQueue.ProcessAcks( ranges );  
        ...  
  
        BeginTransaction (NULL,  
            ISOLATIONLEVEL_SERIALIZABLE, 0, pITxOptions,  
            &pITransaction);  
        ...  
  
        return Greeting;  
    }  
}
```

10 lines
security

1,804 lines
reliable messaging

25,507 lines
transactions

Total lines 27,321



Productivity

Using *Visual Studio 2005* and *WCF*

```
[ServiceContract]
class HelloService
{
    [OperationContract]
    [PrincipalPermission(SecurityAction.Demand,
        Role = "Administrators")]
    [TransactionFlow(TransactionFlowOption.Mandatory)]
    [OperationBehavior(TransactionScopeRequired = true,
        TransactionAutoComplete = true)]
    String Hello(String Greeting) {return Greeting;}
}
```

```
<bindings>
  <wsHttpBinding>
    <binding name="Binding1" transactionFlow="true">
      <security mode="Message">
        <message clientCredentialType="windows"/>
      </security>
      <reliableSession enabled="true" />
    </binding>
  </wsHttpBinding>
</bindings>
```

1 code + 2 config (elem.)
security

0 code + 1 config (elem.)
reliable messaging

2 code + 1 config (attr.)
transactions

Total lines:
code (what) + 4 config (how)

Getting Started with WCF

demo

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Questions...